

**IMPLEMENTATION OF V- SHAPE MODEL METHOD IN
DEVELOPMENT OF INVENTORY MANAGEMENT SYSTEM**



Lecturer:

Dimas Novian Aditia Syahputra, S.Tr.T., M.Tr.T.

Prepared by:

Aisyah Currents	25091397108
Alya Nurin Fadilah	25091397097
Sayyidah Farah Amellia	25091397035
Soraya Ikhsania Balqis	25091397076

**INFORMATIC MANAGEMENT MAJOR
FACULTY OF VOCATION
UNIVERSITAS NEGERI SURABAYA**

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CHAPTER I

INTRODUCTION

1.1 Background of the Paper

Rapid advancements in information technology have brought about significant transformations in Indonesia's retail sector. Web-based commerce, or e-commerce, has become the backbone of the digital economy, which continues to grow year after year. According to official data from Bank Indonesia, the value of national e-commerce transactions has increased consistently from Rp 266.3 trillion in 2020 to Rp 476.3 trillion in 2022, with the highest surge of 50.7% occurring in 2021 (Kontan Data Center, 2024). This growth trend reflects a shift in consumer behavior, with people increasingly relying on digital platforms to meet their daily needs, while also opening up significant business opportunities for entrepreneurs across various market segments.

The Central Statistics Agency (BPS, 2023), in its 2022/2023 E-Commerce Statistics publication, noted that the number of e-commerce businesses in Indonesia is estimated to have reached 2,995,986 in 2022, a 4.46% increase from the previous year. BPS also reported that 9 out of 10 e-commerce businesses use instant messaging as their primary sales and promotional channel. This situation indicates increasingly intense competition in the digital commerce sector, making efficient and computerized business operations an urgent necessity for every business operator to survive and thrive.

Among the various operational aspects of e-commerce, inventory management is one of the most critical elements that directly impacts the smooth operation of services and the company's profitability. The challenges of inventory management in an e-commerce environment are far more complex than in conventional retail, given the high volume of transactions, the wide variety of products, and the demands for fast and accurate order fulfillment. Herlambang and Rachmadi (2020) emphasize that information systems developed to support operational management must be capable of providing real-time information to minimize data errors and improve the overall efficiency of business processes.

The most common problems faced by web-based e-commerce businesses are inaccurate inventory data, overstocking and stockouts, as well as delays in the procurement process. Kurniawati and Ikhwan (2023) demonstrated in their research that the absence of a structured inventory management system leads to inefficient stock control processes, a lack of data integration across departments, and the inability to generate inventory reports automatically. These conditions ultimately result in a decline in the quality of customer service and the potential for financial losses for the company.

To address these issues, the development of a web-based inventory management system is a strategic solution that needs to be implemented. Nasution,

Manurung, and Rahayu (2022) demonstrate that the implementation of a web-based information system for monitoring inventory significantly improves data accuracy, speeds up the reporting process, and facilitates procurement decision-making compared to manual recording methods. A computerized system is capable of automatically recording inventory, providing notifications when stock levels approach the minimum threshold, and generating comprehensive reports that can be accessed anytime and from anywhere.

The success of information system development is not only determined by the technical features it offers but also heavily depends on the selection of the appropriate Software Development Life Cycle (SDLC) model. Balaji and Murugaiyan (2012) state in their comparative study that each SDLC model has its own characteristics, advantages, and limitations; therefore, its selection must be tailored to the nature and complexity of the project at hand. Commonly used models include Waterfall, Agile, Spiral, Prototyping, and V-Shape, each with a different approach to managing the software development cycle.

The V-Shape Method (V-Model) is an extension of the Waterfall model that specifically emphasizes verification and validation at every stage of development. Herlambang and Rachmadi (2020) explain that the V -Model is shaped like the letter “V,” where the left side represents the development phases—ranging from requirements analysis, system design, detailed design, to implementation—while the right side represents the corresponding testing phases, namely unit testing, integration testing, system testing, and User Acceptance Testing (UAT). Based on the principle that each development phase has a corresponding testing phase, the V-Model ensures system quality from the early stages through continuous validation before implementation takes place.

Wu (2024), in his exploratory study on the application of the V -Model from a systems engineering perspective, presented at the IEEE/ACM Conference on AI Engineering (CAIN 2024), concludes that the V-Model has advantages in terms of clarity of system boundaries, consistency in validation and verification (V&V), and the ability to manage interdisciplinary collaboration in the development of complex systems. The V-Model’s primary strength lies in its ability to detect and correct errors earlier in the development cycle, thereby significantly reducing correction costs, which tend to rise if errors are discovered in the final stages or after the system is launched.

The application of the V-Model in the development of an inventory management system for a web-based trading application is considered highly relevant given the high demands for system reliability and accuracy in managing inventory data in real time. Errors in the inventory management system can directly result in business losses in the form of lost sales due to stockouts or wasted capital due to overstocking. Therefore, a quality-oriented and systematic testing approach such as the V-Model is highly suitable for implementation.

1.2 Problem of the Paper

- A. What are the common issues in manual inventory data management for online retail shops, and how does this system address them?
- B. How are the verification and validation phases of the V-Shape Model implemented to ensure that the inventory management system features meet user requirements?
- C. How can sales transaction recording be integrated into the inventory system to ensure real-time stock updates?

1.3 Purpose of the Paper

- A. To identify common issues in manual inventory management for online shops and develop a system that effectively addresses these challenges.
- B. To implement the verification and validation phases of the V-Shape Model to ensure the system features align with user requirements.
- C. To integrate sales transaction recording into the inventory system for accurate, real-time stock updates.

1.4 Benefits of the Paper

- A. For the Shop Owner: Provides an automated and organized system to reduce human error in stock counting and financial recording.
- B. For the Customers: Ensures accurate information regarding item availability, leading to a better shopping experience.
- C. For Academics: Serves as a reference for implementing the V-Shape Model in software engineering projects, specifically within the e-commerce and inventory sector.

CHAPTER II

THEORY AND DISCUSSION

2.1 Software Engineering

Software Engineering is a planned method for designing, building, testing, and maintaining software. This process includes the use of principles, practices, methods, and tools that have been proven effective in developing quality software solutions (Mulyani, Tiawan, and Nugraha, 2021). Software engineering is a field of science and practice concerned with the process of creating effective, efficient, and reliable software. The goal is to develop software that meets user needs, while maintaining optimal quality, security, scalability, and performance.

Software engineering encompasses user needs analysis. This involves identifying the functional and non-functional requirements of the system to be developed. These requirements include the desired features and functions of the software, as well as the constraints and requirements that must be met. After understanding user needs, the next step is to design the software structure. Software architecture describes the overall structure of the system, including elements, modules, and the relationships between these components (Sulistiyorini, 2009). This design ensures that software can be developed in a structured and organized manner.

Once the design is complete, software engineering involves actually building the software. This process includes writing code, testing functionality, and ensuring the software functions as intended. The development team will collaborate to divide tasks, implement good code standards, and utilize appropriate development tools. Throughout the software development process, software engineering encourages the implementation of configuration management practices. These activities include monitoring changes to the source code, ensuring consistent versions are deployed, and maintaining revision records. These activities facilitate the development team's collaboration and rapid resolution of issues as they arise.

Furthermore, software engineering also includes comprehensive software testing (Huda et al., 2022). This includes functional testing, unit testing, integration testing, and performance testing. The goal is to ensure that the software operates as expected and does not contain any exploitable defects or vulnerabilities. During the testing phase, software engineering also involves finding and fixing any errors that have been detected. The development team utilizes bug tracking tools to detect and resolve issues. This process is repeated repeatedly until the software is deemed sufficiently stable and flawless.

After a program is created and tested, software engineering involves deploying and installing the software in a designated environment (Toba et al., 2022). These activities include server management, network configuration, and user training. This process ensures that the software is ready for end-user use. Furthermore, software engineering also encompasses the maintenance and upkeep

of existing software. This includes troubleshooting, strengthening security, fixing errors, and improving efficiency. Effective software maintenance is crucial for maintaining the software's performance over time and meeting evolving needs. Throughout the entire software lifecycle, software engineering also focuses on project management. This includes project planning, resource allocation, and monitoring project progress. Effective project management ensures that projects are completed on schedule, within budget, and achieve predetermined objectives.

Finally, software engineering encompasses professional ethics and responsibility. Software developers are expected to respect user privacy, safeguard sensitive data, and adhere to ethical standards dictated by their profession. They are expected to continuously hone their skills through training and self-development (Muqorobin, 2023). In general, software engineering is a scientific and practical method for creating quality software. With an organized approach and proven methods, software engineering ensures that software meets user expectations, functions optimally, and is managed efficiently throughout its lifespan (Irfan Sadida1, M. Gunawan Setyadi, 2022).

2.2 V-Shape Model

The V- shape model is often referred to as an evolution of the waterfall model, as the V-model features similar phases. The difference lies in the process flow: while the waterfall model follows a linear path, the V-model features a branching flow that connects each development phase with the corresponding testing phase. This model is designed to simplify understanding of the complexity of system development and to provide a systematic procedure for project or product development. The name “V-model” itself derives from the terms ‘verification’ and “validation,” which emphasize the process of examining and testing the system to ensure it meets the specified requirements (Yuningsih, 2022).

Model V (Validation/Verification Model) is one of the methods in the System Development Life Cycle (SDLC) that illustrates the relationship between the system development phases and their corresponding testing phases. In this model, the testing process is conducted in a more structured and detailed manner because each development phase has a corresponding testing phase. The development phases include requirement analysis, requirement specification, design specification, and program specification, while the testing phases include acceptance testing, system testing, integration testing, and unit testing (Yuningsih, 2022).

2.3 Inventory Management System in the Clothes shop

Inventory Management System (IMS) is a combination of technology and processes used to monitor and manage a company's stocked goods. In the context of a clothes shop, an IMS tracks every item from the moment it is received into the

warehouse or store until it is sold to the customer. This system is crucial for maintaining the balance between having enough stock to meet customer demand and avoiding the costs associated with excess inventory.

According to Muller (2019), effective inventory management involves controlling stock levels, tracking sales, and managing replenishment. For a clothes shop, especially one operating in the thrift or online sector, inventory management has unique challenges, such as:

- A. Stock Tracking: Each item in a thrift shop is often unique (one-of-a-kind). An IMS helps ensure that once an item is sold, it is immediately removed from the digital catalog to prevent "double selling."
- B. Real-time Updates: Integrating the system with sales transactions ensures that the stock count is always accurate, allowing the owner to make informed decisions about restocking or pricing strategies.
- C. Efficiency: By replacing manual recording with an automated system, the shop can reduce human errors, save time, and provide better data for financial reporting.

In this project, the inventory management system is designed to provide a centralized database that stores detailed information about each clothing item, including its category, price, and status, ensuring a smooth operational flow for the business.

CHAPTER III RESEARCH METHODOLOGY

3.1 Planning

The planning stage is the initial phase in the development of the inventory management system for a clothes shop. At this stage, the development team identifies several problems commonly faced by clothes shop owners, such as manual stock recording, inaccurate inventory data, difficulties in monitoring sales transactions, and the lack of integrated sales reports. These problems may reduce operational efficiency and increase the possibility of human error in managing store activities. The overall planning process carried out in this stage is illustrated in the following flowchart.

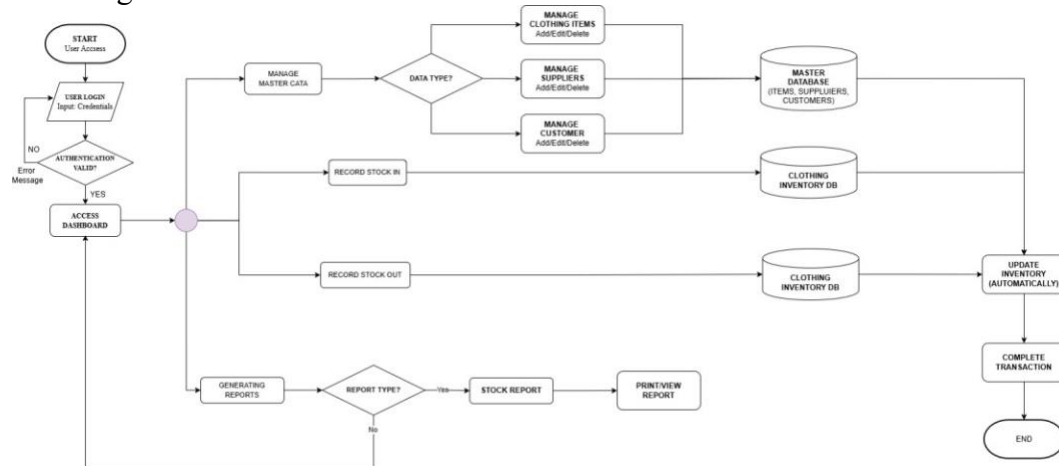


Figure 1. Flowchart

After identifying the problems, the team determines the objectives of the system development. The main objective of this project is to develop a web-based inventory management system that can help clothes shop owners manage product data, monitor stock availability, record sales transactions, and generate reports automatically and accurately. The system is expected to improve efficiency, minimize recording errors, and support business operations in a more organized manner.

In addition, the development team conducts literature studies related to software engineering, inventory management systems, clothes shop business operations, and the V-Shape Model development method. This activity aims to obtain theoretical references and understand suitable approaches for developing the proposed system. Furthermore, the team determines the hardware and software requirements needed for system development, including programming tools, database systems, and supporting technologies.

3.2 Requirement Analysis

The requirement analysis stage aims to identify and define user and system requirements to ensure that the developed application functions according to user needs. In this project, the analysis process is conducted by observing inventory management activities in clothes shops and identifying problems that commonly occur in manual systems. Based on the analysis results, the system requires several functional and non-functional requirements.

1. Functional Requirements

- Allow administrators to log into the system securely.
- Manage inventory data, including adding, editing, deleting, and viewing product information.
- Record sales transactions automatically.
- Update stock quantities in real time after transactions occur.
- Generate inventory and sales reports.
- Search for products quickly based on categories or product names.

2. Non-Functional Requirements

- Be accessible through a web browser with an internet connection.
- Provide a user-friendly and responsive interface.
- Ensure data security through authentication and access control.
- Maintain stable system performance during operation.
- Use a database management system to store and organize data efficiently.

The results of this requirement analysis become the foundation for the next stages, including system modeling, interface design, implementation, and testing within the V-Shape Model development process.

3.3 System Modeling

The system modeling stage is carried out to describe the structure, workflow, and behavior of the inventory management system before implementation. System modeling helps developers understand the relationship between system components and ensures that the developed system meets user requirements. In this project, the Unified Modeling Language (UML) is used as the main modeling approach, including Use Case Diagram, Class Diagram, and Sequence Diagram.

The Use Case Diagram is used to describe the functional requirements of the system and the interaction between users and the system. In this project, the main actor is the administrator who can perform activities such as managing clothing item data, supplier data, customer data, recording stock in and stock out transactions, and generating reports.

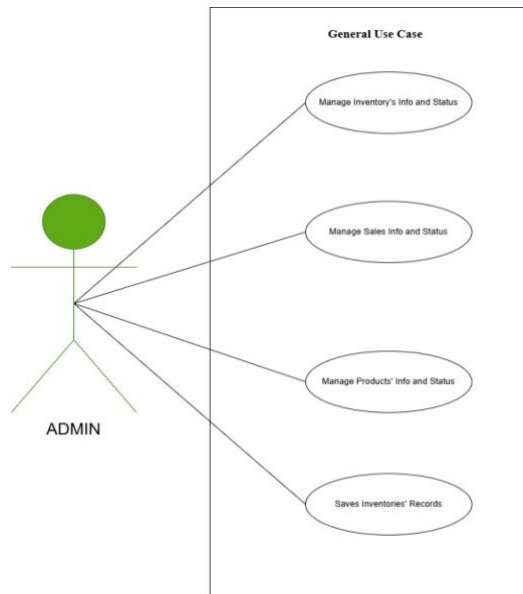


Figure 2. UML Use Case Diagram

The Class Diagram is used as a structural representation of the system. It illustrates the classes, attributes, methods, and relationships between objects within the inventory management system. The Class Diagram helps developers design a well-structured system architecture and database implementation.

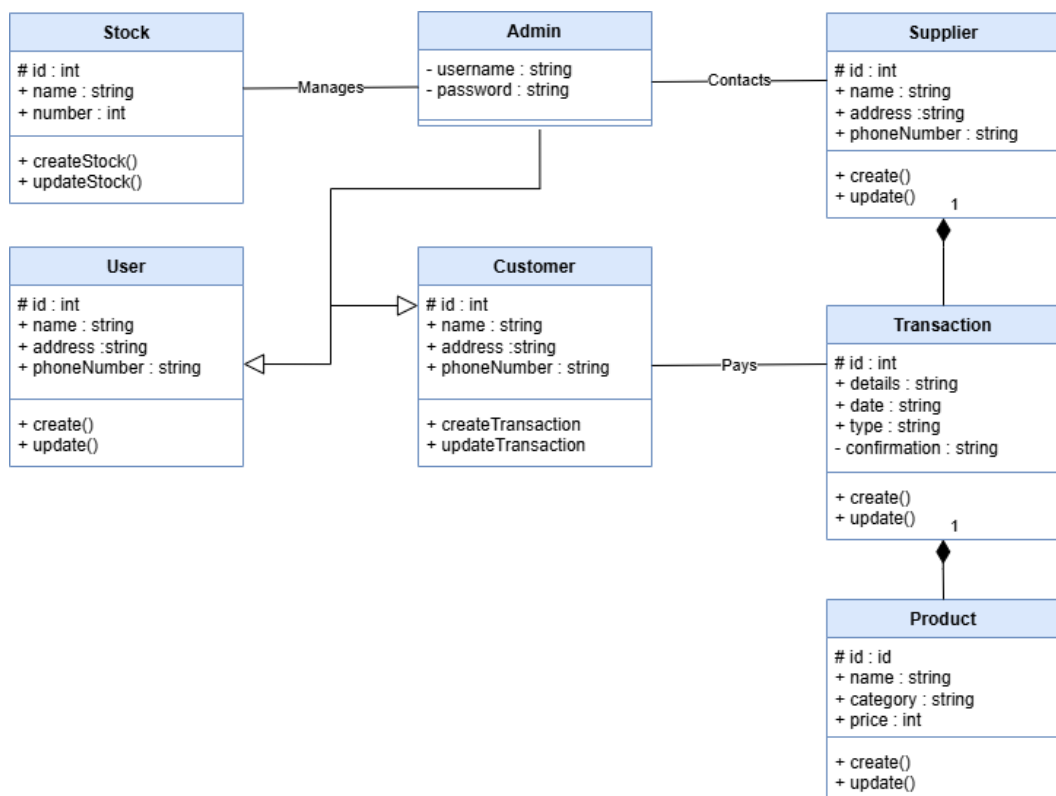


Figure 3. UML Class Diagram

The Sequence Diagram is used for behavior analysis by illustrating the sequence of interactions between users and system components during a specific process. This diagram describes how messages and data are exchanged within the system, such as during login, stock transactions, and report generation processes.

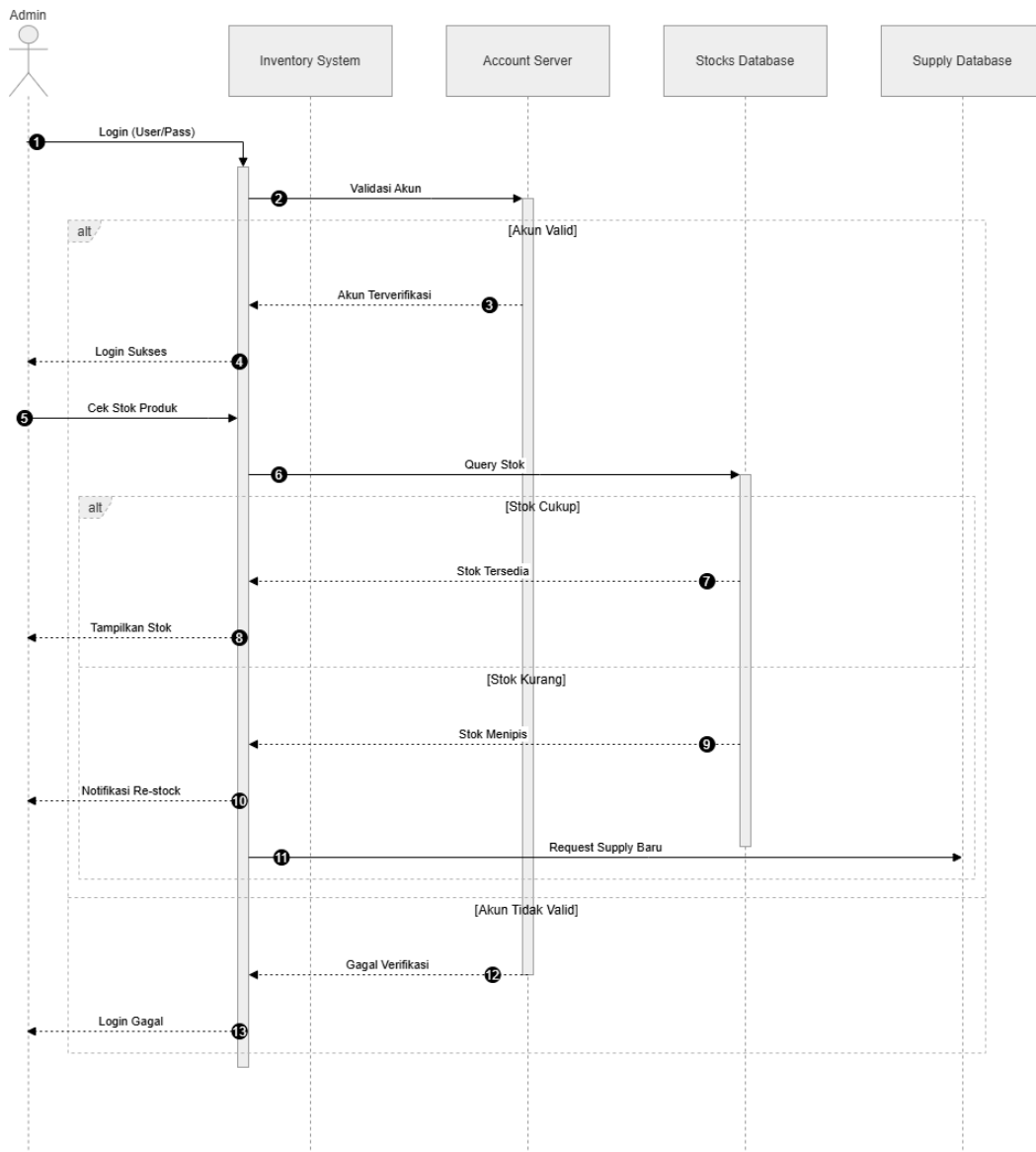


Figure 4. UML Sequence Diagram

Through UML-based system modeling, the development process can be conducted in a more organized and systematic manner, reducing errors during the implementation stage and improving system design quality.

3.4 Interface Design

The interface design stage focuses on designing the appearance and interaction of the inventory management system to ensure that the system is easy to

use and visually understandable for users. A good interface design can improve user experience and help administrators perform tasks more efficiently.

In this project, the interface design includes several main pages such as the login page and dashboard page. The login page is designed to provide secure access for administrators by requiring a username and password before entering the system. Meanwhile, the dashboard page displays important information such as stock summaries, recent transactions, and quick access to the main features of the application, allowing users to monitor system activities efficiently.

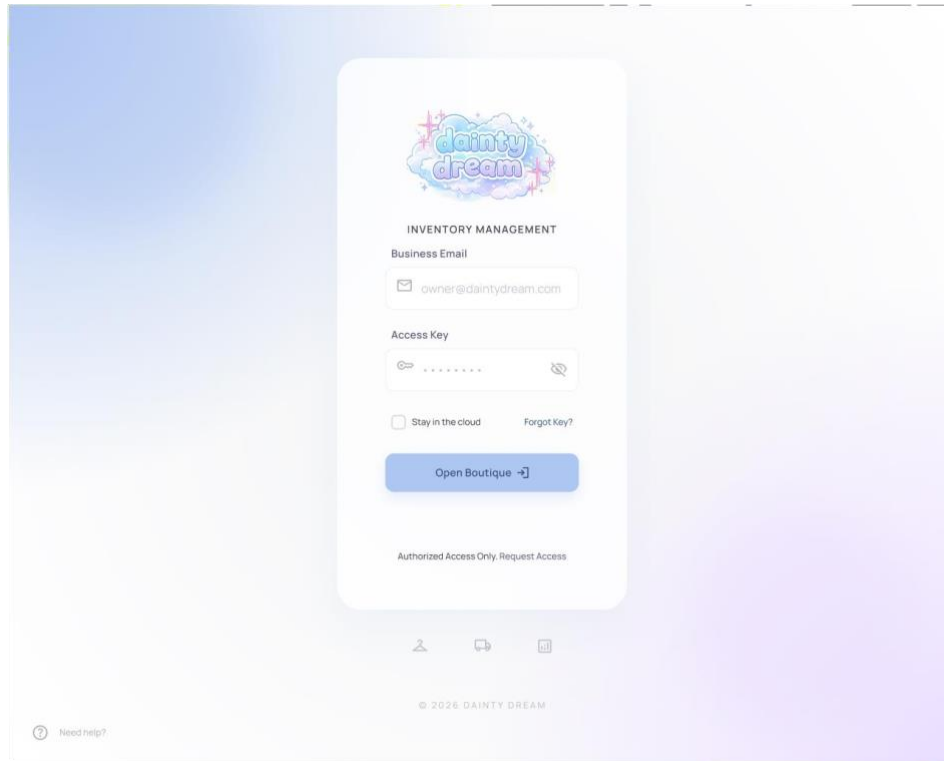


Figure 5. Login Page

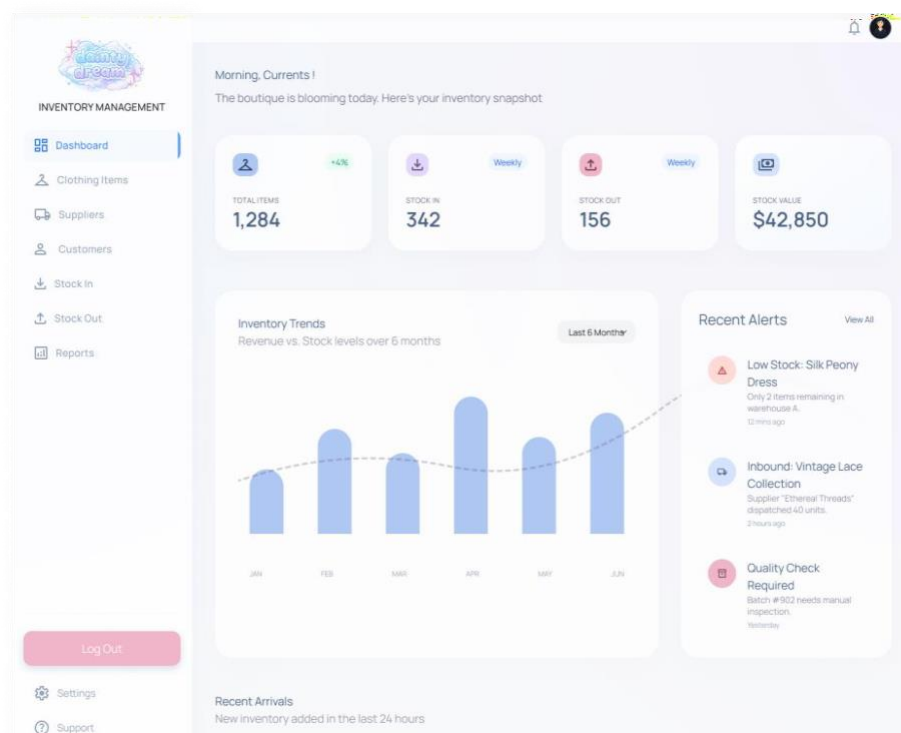


Figure 6. Dashboard Page

The master data pages in this project consist of the clothing item page, supplier page, and customer page. The clothing item page is designed to manage product information such as clothing names, categories, prices, and stock quantities. The supplier page allows administrators to store and manage supplier information, including supplier names, contact details, and addresses. Meanwhile, the customer page is used to manage customer data, making it easier to record customer information and support activities effectively.

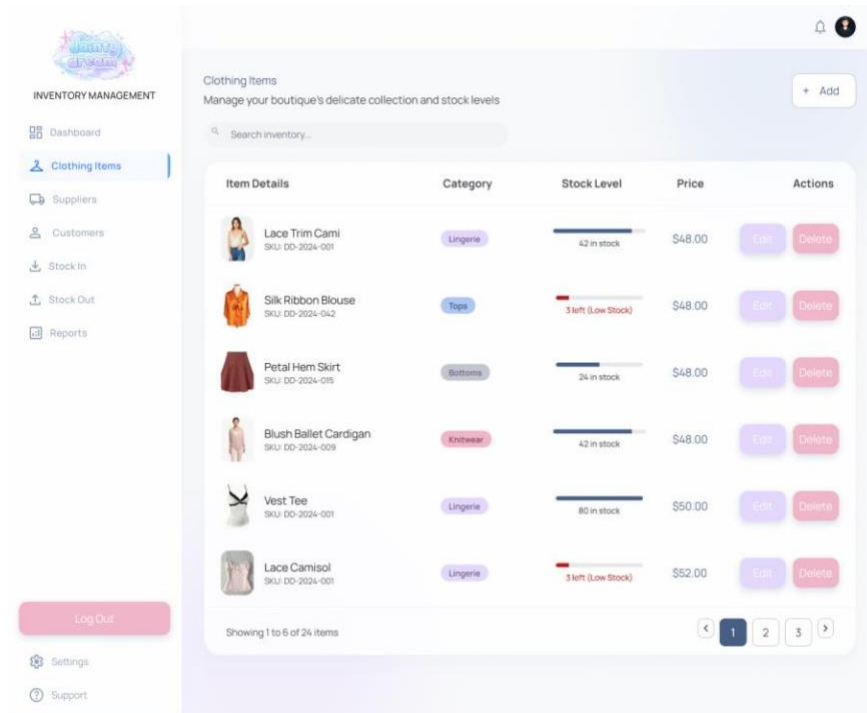


Figure 7. Clothing Items Page

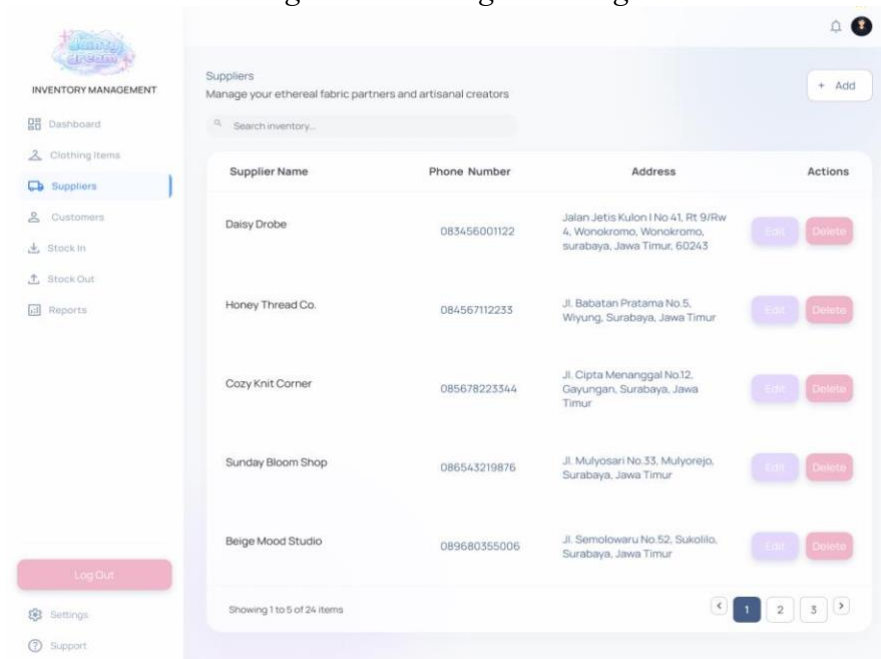


Figure 8. Supplier Page

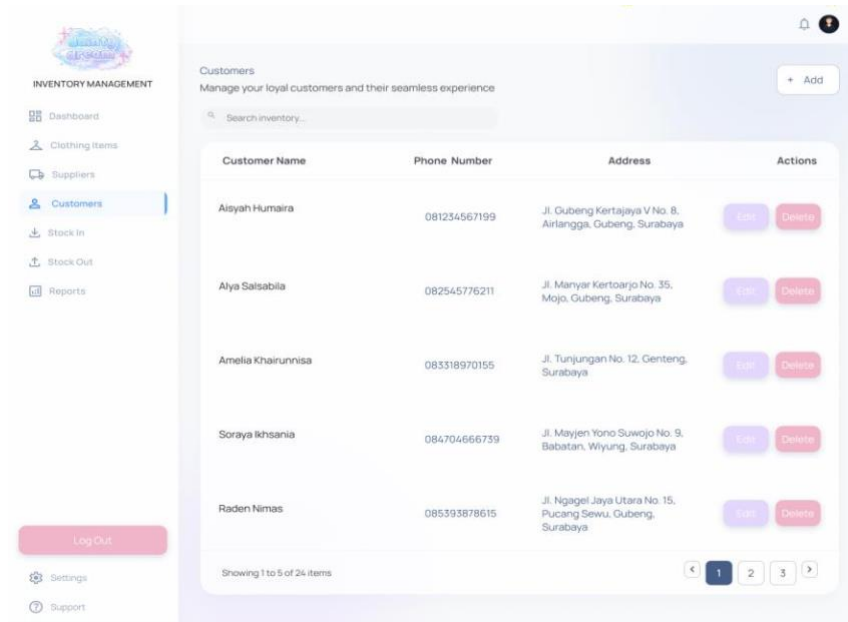


Figure 9. Customer Page

The stock in page is designed to record incoming products from suppliers and automatically increase the available stock quantity in the inventory system. Meanwhile, the stock out page is used to record outgoing products or sold items, helping administrators monitor stock reductions accurately in real time. The report page provides stock in and stock out reports that help store owners monitor inventory movement and manage business activities effectively.

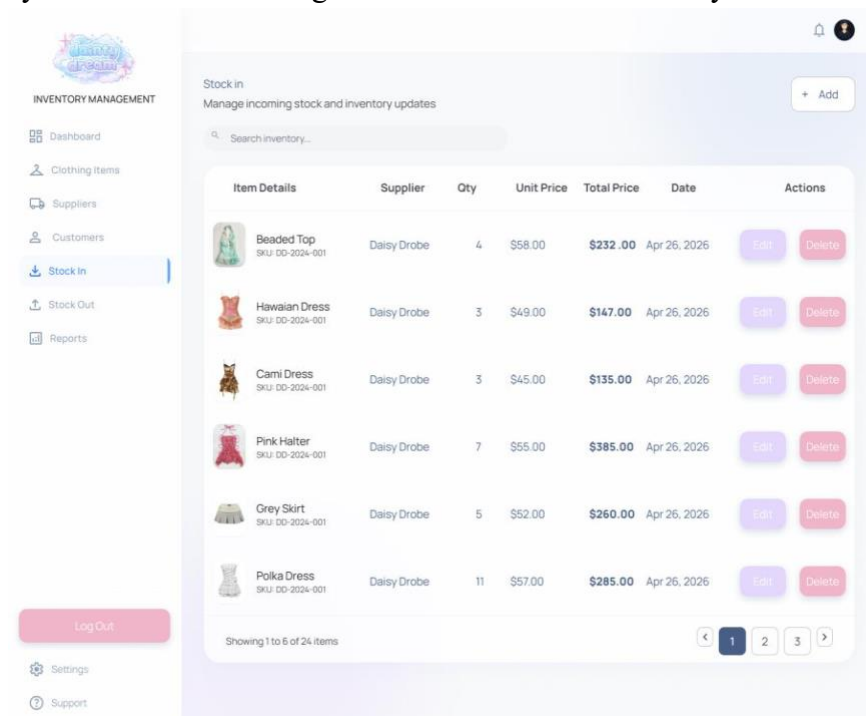


Figure 10. Stock In Page

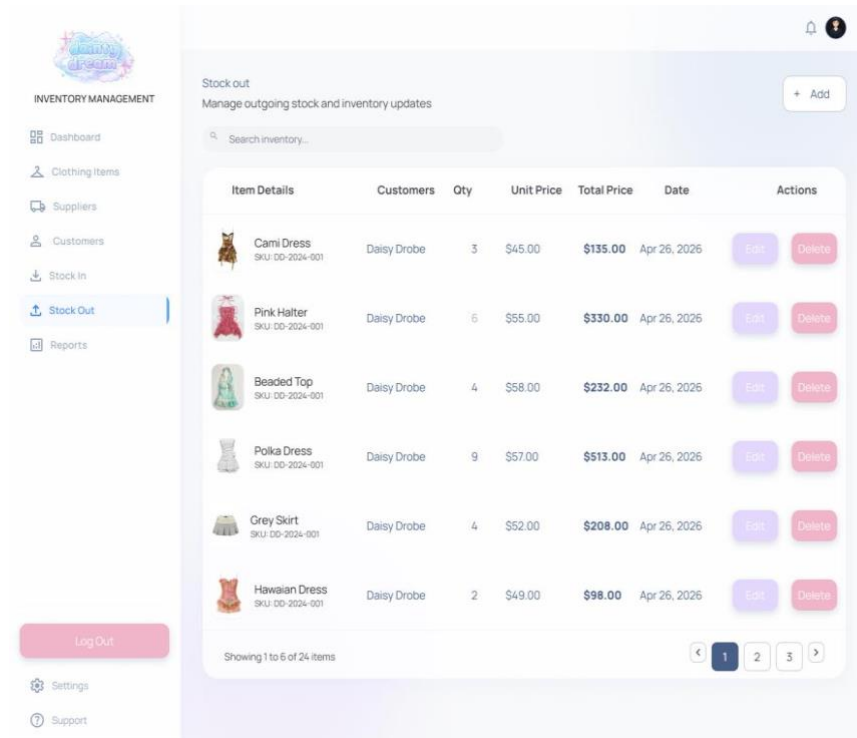


Figure 11. Stock Out Page

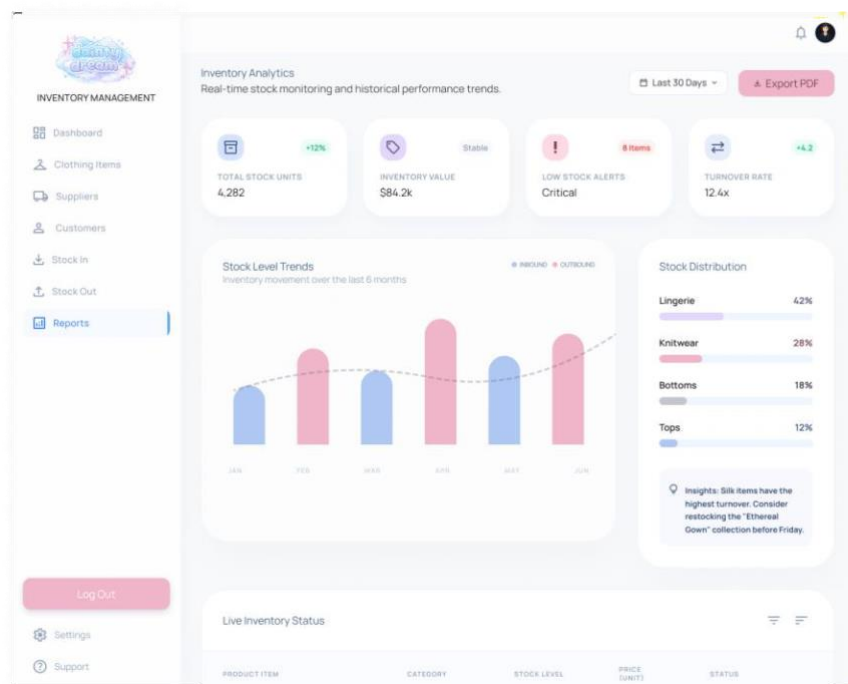


Figure 12. Report Page

The interface design process also considers user-friendly principles, responsive layouts, clear navigation, and consistency in visual appearance. Mockups and interface prototypes are created before implementation to ensure that the designed interface meets user needs and system requirements.

3.5 Implementation

The implementation stage is the process of developing the inventory management system based on the designs and requirements that have been prepared in the previous stages. At this stage, the system features, database structure, and user interface are translated into program code using selected technologies and development tools.

The system is developed as a web-based application to allow users to access the system more easily through a browser. The implementation process includes creating the database, coding system functionalities, integrating the database with the user interface, and configuring the application environment. The developed features include product management, inventory monitoring, sales transaction recording, and report generation.

In this project, the implementation process uses programming languages, database systems, and supporting technologies suitable for web application development. The implementation stage aims to ensure that all designed features can function properly according to user requirements and system specifications.

3.6 Testing Method

The testing stage is conducted to ensure that the inventory management system functions correctly and meets user requirements. In this project, the testing process follows the V-Shape Model approach, where each development stage is associated with a corresponding testing activity.

Several testing methods are carried out during this stage. Unit testing is used to test individual components or modules of the system to ensure that each feature works properly. Integration testing is performed to verify that all modules can communicate and function together correctly within the system.

System testing is conducted to evaluate the overall performance and functionality of the application. In addition, User Acceptance Testing (UAT) is performed to determine whether the system meets user expectations and business needs. Through these testing activities, errors and system weaknesses can be identified and corrected before the system is fully deployed.

3.7 Deployment

The deployment stage is the process of preparing and releasing the inventory management system for actual use. At this stage, the application and database are configured and installed in the target environment so that the system can be accessed and used by users.

The deployment process includes setting up the web server, importing the database, configuring system settings, and ensuring that all system features function properly in the production environment. The system is then tested again to confirm that it operates smoothly after deployment.

This stage also includes providing guidance for users regarding how to operate the system effectively. Proper deployment ensures that the inventory management system can support daily operational activities efficiently and reliably.

3.8 Maintenance

The maintenance stage is carried out after the system has been deployed and used by users. The purpose of maintenance is to ensure that the system continues to function properly, remains secure, and can adapt to future needs or changes.

Maintenance activities include fixing system errors or bugs, updating system features, improving system performance, and maintaining database security. In addition, regular monitoring is performed to identify potential problems and ensure that the application operates efficiently.

Through continuous maintenance, the inventory management system can provide stable performance, support business activities effectively, and remain relevant to user requirements over time.

CHAPTER IV CONCLUSION

4.1 Conclusion

Based on the results of the research and development conducted, it can be concluded that the implementation of an Inventory Management System in online thrift shop businesses has successfully provided an effective digital solution to overcome the challenges of conventional stock management. The transition from manual recording methods to a computerized system has been proven to significantly improve operational efficiency and inventory data accuracy, while minimizing the risk of human error in recording incoming and outgoing goods. The application of the V-Shape Model methodology in the system development cycle ensures that every feature, from master data management to transaction reporting, has undergone rigorous verification and validation stages, ensuring that the system's functions are fully aligned with the needs of users and business owners.

Furthermore, the integration of sales transaction recording directly linked to the database allows for automatic, real-time stock updates. This not only effectively prevents double selling of unique items but also facilitates business owners in generating credible periodic reports to support future strategic decision-making. Thus, this system is expected to increase the competitiveness of thrift shop businesses in the rapidly evolving digital commerce ecosystem.

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